

Algorithmic fairness

Lecture 1 – Introduction and formalism

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Artificial Intelligence

Supervised Learning:

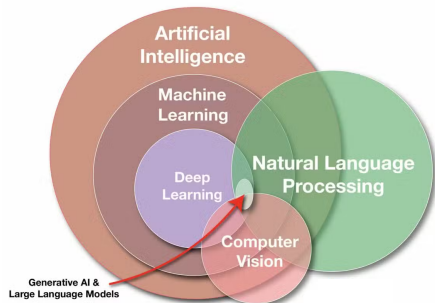
Data: Sample of size n

$$\underbrace{(\text{feature}, \text{label})}_{\mathbf{X}_i \in \mathcal{X} \quad Y_i \in \mathcal{Y}} \sim \mathbb{P}$$

Predictions: $f : \mathcal{X} \rightarrow \mathcal{Y}$

Training: $\ell : \mathcal{Y} \times \mathcal{Y} \mapsto \mathbb{R}_+$

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \ell(Y_i, f(\mathbf{X}_i))$$



Main frameworks:

Regression $\mathcal{Y} \subset \mathbb{R}$

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (Y_i - f(\mathbf{X}_i))^2$$

General use: Automatic decision based on $\hat{f}$

Classification $\mathcal{Y} := \{0, 1\}$

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{Y_i \neq f(\mathbf{X}_i)}$$

Classical approach in Statistical learning

Bayes optimal: $f^* \in \arg \min_f \mathcal{R}(f) := \mathbb{E} \ell(Y, f(\mathbf{Z}))$

Estimator: $\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \ell(Y_i, f(\mathbf{Z}_i)) (+ \text{penalty}(f) \text{ eventually})$

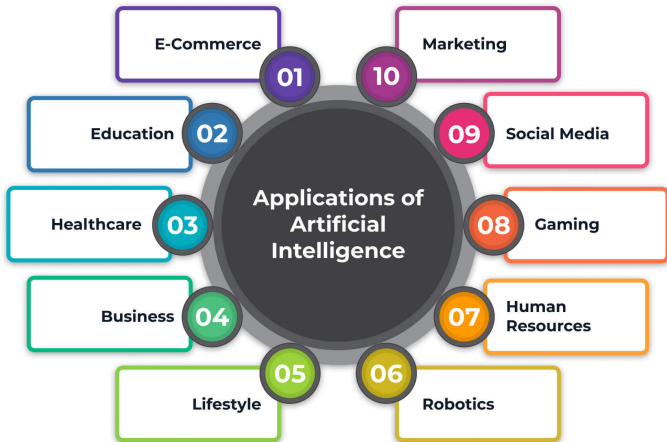
Excess Risk: $\mathcal{E}(\hat{f}) = \mathcal{R}(\hat{f}) - \mathcal{R}(f^*)$

From the statistical point of view, we look for algorithms such that (with high probability / in expectation)

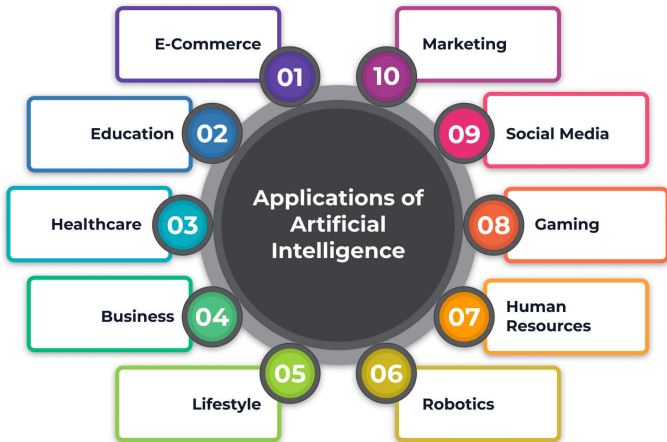
$$\mathcal{E}(\hat{f}) \leq \varepsilon$$

for some small ε – usually depends on n .

AI with success



AI with success



But also malicious mistakes...

Historical example: COMPAS dataset

Correctional Offender Management Profiling for Alternative Sanctions

- ▶ Risk-assessment software developed and owned by **Northpointe**
- ▶ The software is used in U.S. courts to predict recidivism risks of defendants
- ▶ Investigation of **Propublica**

“Overall, Northpointe’s assessment tool correctly predicts recidivism 61 percent of the time. But blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend. It makes the opposite mistake among whites: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes.”

- ▶ https://fairlearn.org/v0.4.6/auto-examples/plot_binary_classification_COMPAS.html
- ▶ <https://www.propublica.org/datastore/dataset/compas-recidivism-risk-score-data-and-analysis>

Motivating examples: Google translate



Machine learning



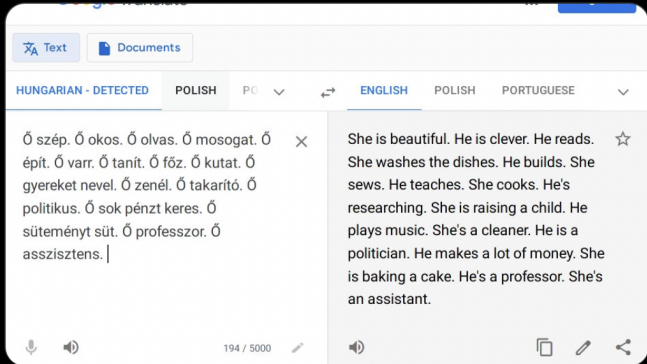
Randy Olson @randal_olson · 4h

...

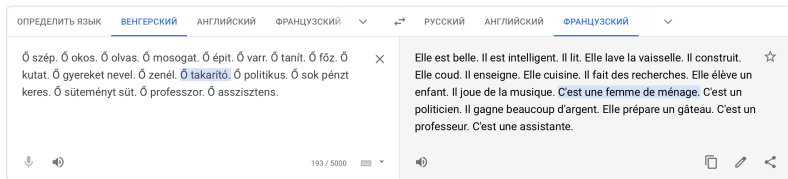
Hungarian has no gendered pronouns, so Google Translate makes some assumptions.

#CodedBias in Google Translate. #DataScience #MachineLearning

Source: reddit.com/r/europe/comme...



Motivating examples: Google translate



Отправить отзыв

Motivating examples: Twitter cropping

Fact: Twitter automatically **crops large** images in order to fit the size of an average mobile screen.

Original



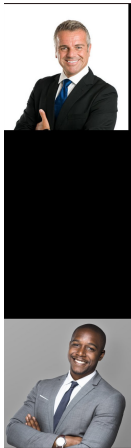
Cropped



Motivating examples: Twitter cropping

Fact: Twitter automatically **crops large** images in order to fit the size of **an average mobile screen**.

Question: How will Twitter crop these two images??



We see two men in fancy suits, one of them has a fancy watch – clearly **successful** guys.

The only difference is the **color of their skin** and the **relative position**.

Motivating examples: Twitter cropping

Fact: Twitter automatically **crops large** images in order to fit the size of an average mobile screen.

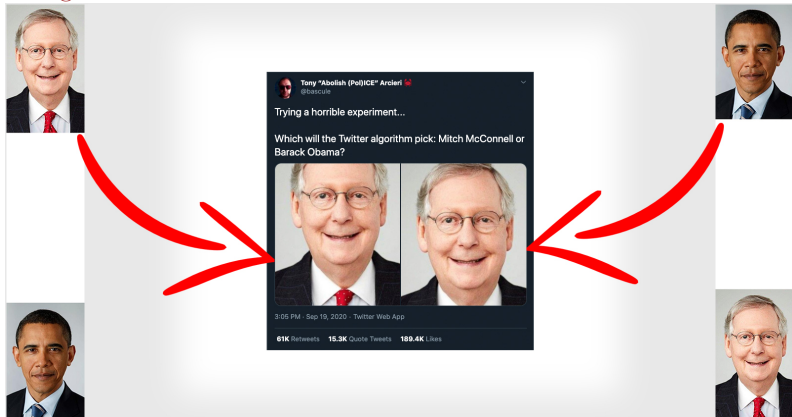


Disclaimer: Twitter have acknowledged the problem and have publicly apologized

(https://blog.twitter.com/en_us/topics/product/2020/transparency-image-cropping.html)

Motivating examples: Twitter cropping

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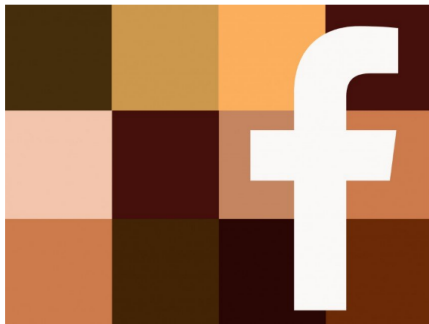
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Now we can crop it ourselves.

(https://blog.twitter.com/en_us/topics/product/2020/transparency-image-cropping.html)

Motivating examples: Facebook ads

MIT
Technology
Review



MS. TECH/ LOGO: FACEBOOK

Artificial Intelligence / Machine Learning

Facebook's ad-serving algorithm discriminates by gender and race

Even if an advertiser is well-intentioned, the algorithm still prefers certain groups of people over others.

by Karen Hao

Apr 5, 2019

Another example: modern redlining

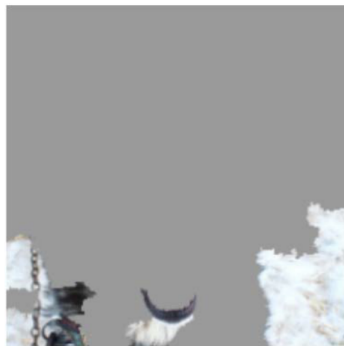
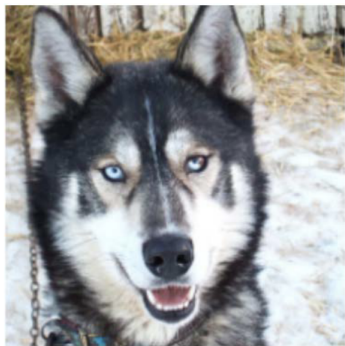
► Wikipedia

“Redlining is a discriminatory practice in which services (financial and otherwise) are withheld from potential customers who reside in neighborhoods classified as “hazardous” to investment; these neighborhoods have significant numbers of racial and ethnic minorities, and low-income residents.”

► Barocas et al., *Fairness and Machine Learning*, (2019)

“ Amazon uses a data-driven system to determine the neighbourhoods in which to offer free same-day delivery. A 2016 study found stark disparities in the demographic make-up of these neighbourhoods: in many U.S. cities, white residents were more than twice as likely as black residents to live in one of the qualifying neighbourhoods.”

Husky/Wolf classification

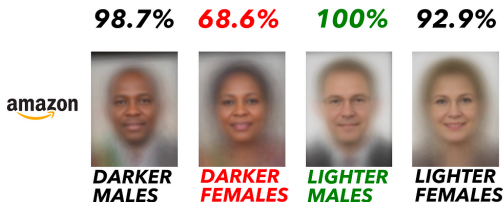


- A husky (left) is misclassified as a wolf because the pixels deemed characteristic of wolves (right) in fact correspond to the snowy background.

And many other Fairness Failures...

- ▶ **Amazon's Hiring Algorithm (2018):**
 - ▶ Discriminated against women in tech roles due to biased training data.
- ▶ **Apple Card Credit Limits (2019):**
 - ▶ Allegations of gender bias in determining credit limits.
- ▶ **Facial Recognition Bias:**
 - ▶ Higher error rates for people with darker skin tones.

August 2018 Accuracy on Facial Analysis Pilot Parliaments Benchmark



Amazon Rekognition Performance on Gender Classification

A need for regulation (at the level of EU)

Context

- ▶ The rise of digital technologies impacts consumer rights
- ▶ The EU evaluates existing consumer protection laws for digital fairness
- ▶ Goal: Ensure fair, transparent, and equitable practices for consumers
- ▶ Example: Misuse of consumer data and opaque pricing algorithms

Objectives of the Study

- ▶ Assess effectiveness and relevance of current laws
- ▶ Ensure coherence and added value at the EU level
- ▶ Identify gaps and areas needing reinforcement
- ▶ Example: Addressing dark patterns that mislead consumers

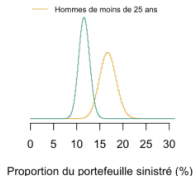
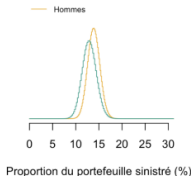
Insurance, the US case

Importance to be aware of the legislation of the country/city:

↪ US **car insurance** protected variables

	CA	HI	GA	NC	NY	MA	PA	FL	TX
Genre	×	×	•	×	•	×	×	•	•
Âge	×	×	•	×	•	×	•	•	•
Expérience de conduite	•	×	•	•	•	•	•	•	•
Antécédents de crédit	×	×	•	•	•	×	•*	•	•
Éducation	×	×	×	×	×	×	•	•	•
Profession	×	×	×	•	×	×	•	•	•
Situation d'emploi	×	×	×	•	×	×	•	•	•
Situation de famille	•	×	•	•	•	×	•	•	•
Situation résidentielle	×	×	•	•	•	×	•	•	•
Adresse/code postal	•	•	•	•	•	•	•	•	•
Antécédents d'assurance	•	•	•	•	•	•	•	•	•

In France/EU: need to prove a causal link to be able to use a sensitive variable (age, sexe for instance)



Applications in insurance

General goal

Built accurate pricing without being too intrusive in insured persons life and that respects common ethic considerations.

Some examples

- ▶ **Contracts pricing:** life/health/home/car insurance
- ▶ **Reparation costs** from pictures of a damaged car
- ▶ **Actuarial risk** of individuals
- ▶ **Fraud detection**

Commonly used sensitive variables: sex - age - name - ethnic/parents origin - salary - family situation - socioeconomic status - address - pregnancy/motherhood - healthy or sick - smoker or not – facial image - ...

Question

How should we take into account fairness considerations and build accurate pricing in the same time?

Key Themes

- ▶ **Dark Patterns:** Interface designs that mislead users
 - ▶ Examples: Hidden fees, misleading buttons, forced continuity subscriptions
 - ▶ Impact: Erodes trust and consumer autonomy
 - ▶ Reference: Article 7 of the EU Unfair Commercial Practices Directive prohibits misleading omissions
- ▶ **Platform Transparency:** Lack of clarity in algorithms and data use
 - ▶ Proposal: Mandatory disclosures on algorithms
 - ▶ Example: The AI Act mandates transparency for high-risk AI systems, including recommendation engines
 - ▶ Aim: Enhance consumer understanding and choice
- ▶ **Consumer Rights in Digital Age:** Adapting laws for digital contexts
 - ▶ Focus: Safeguarding disadvantaged groups
 - ▶ Strategies: Education campaigns and support systems
 - ▶ Example: Regulating AI chatbots to prevent exploitation of less tech-savvy users
 - ▶ Goal: Inclusive consumer protection policies

Evolution of EU Laws on Algorithmic Fairness

At the level of data

► 2018: GDPR (General Data Protection Regulation)

Purpose: Protection of personal data of individuals

- Established principles of lawful, fair, and transparent data processing
- *Data must be processed fairly and transparently* (Article 5(1)(a))
- *Right to explanation for automated decisions* (Article 22)
- Example: Algorithms using personal data must not result in unfair discrimination

► 2024: Digital Services Act (DSA)

Purpose: Regulation of digital services (online platforms, social networks, search engines, marketplaces)

- Imposes transparency obligations on online platforms
- Combatting illegal content (hate speech, disinformation, counterfeiting)
- Example: Disclosure of criteria for ranking content or ads

More specifically on IA

► 2024: AI Act (Artificial Intelligence Act)

Purpose: Regulation specifically for artificial intelligence systems

- First legal framework regulating AI systems in the EU
- Focus on high-risk AI applications like credit scoring and recruitment
- Example: Transparency requirements for recommendation algorithms

Algorithmic Fairness

Definition: Fairness refers to the absence of any form of bias, discrimination, or unfair treatment in decision-making systems

Challenges

- ▶ Mitigate the bias contained in historical data
- ▶ Reduce influence of a **sensitive attributes** in prediction
- ▶ Algorithm should treat people without discrimination based on **sensitive attributes**
- ▶ Here is a short list of pioneer works
 - ▶ T. Calders et al., *Building classifiers with independency constraints*, (2009)
 - ▶ R. Zemel et al., *Learning fair representations*, (2013)
 - ▶ M. Zafar et al., *Fairness beyond disparate treatment & disparate impact: Learning classification without disparate mistreatment*, (2017)
 - ▶ M. Donini et al., *Empirical risk minimization under fairness constraints*, (2018)
 - ▶ A. Agarwal et al., *A reductions approach to fair classification*, (2019)
 - ▶ S. Barocas et al., *Fairness and Machine Learning*, (2019)

Formalization of fairness in supervised learning

(1/2)

Observations

- ▶ Features: $(\mathbf{X}, S)$, output $Y \in \mathcal{Y}$
- ▶ $S \in \mathcal{S}$ sensitive (or protected) attribute

Sensitive attributes

- ▶ Any decisions based on S are undesirable from an ethical or legal perspective
- ▶ French law number 2008-496, Article 1

“Constitue une discrimination directe la situation dans laquelle, sur le fondement de son origine, de son sexe, de sa situation de famille, de sa grossesse, de son apparence physique, de la particulière vulnérabilité résultant de sa situation économique, apparente ou connue de son auteur, de son patronyme, de son lieu de résidence ou de sa domiciliation bancaire, . . . ”

Formalization of fairness in supervised learning

(2/2)

Fairness through awareness

- ▶ Predictor $f \hookrightarrow$ prediction $f(\mathbf{X}, S)$
- ▶ Drawback: prediction relies on S

Fairness through unawareness

- ▶ Awareness may induce direct discrimination
- ▶ Predictor $f \hookrightarrow$ prediction $f(\mathbf{X})$

Disparate learning process

- ▶ S available at the training step but not for the inference (prediction) step

No fairness through unawareness

Remove S does not enforce fairness

- ▶ X and S are not independent (algorithm learns correlations)
- ▶ Indirect discrimination
- ▶ Classic example $\hookrightarrow$ Redlining
- ▶ French law number 2008-496, Article 1

“ Constitue une discrimination indirecte une disposition, un critère ou une pratique neutre en apparence, mais susceptible d’entraîner, pour l’un des motifs mentionnés au premier alinéa, un désavantage particulier pour des personnes par rapport à d’autres personnes, ... ”

Definition of fairness: *group fairness approach*

Definition of fairness: two frameworks

- ▶ **Group fairness** $\hookrightarrow$ defines fairness at population level
- ▶ **Individual fairness**: *similar* people should be treated *similarly*
Example: Personalized recommendations without unjust bias.

C. Dwork et al., *Fairness through awareness*, (2012)

“*treat like cases as like*” ($\leq$ Aristotle)

“*Ensure that similar individuals are treated similarly*”

Definition for group fairness ($\mathbf{W} = (\mathbf{X}, S)$ or $\mathbf{W} = \mathbf{X}$)

- ▶ independence $\hookrightarrow f(\mathbf{W}) \perp\!\!\!\perp S$
- ▶ separation $\hookrightarrow (f(\mathbf{W}) \perp\!\!\!\perp S) \mid Y$
- ▶ sufficiency $\hookrightarrow (Y \perp\!\!\!\perp S) \mid f(\mathbf{W})$

Back to COMPAS example

Northpointe vs. Propublica

Northpointe: *Sufficiency (Statistical Parity)*

- ▶ Certifies that a person predicted as dangerous will certainly recidivate
- ▶ This score is the same across ethnicity (sensitive variable $S \in \{0, 1\}$)

$$\mathbb{P}(Y = 1 \mid f(\mathbf{W}) = 1, S = 1) \approx \mathbb{P}(Y = 1 \mid f(\mathbf{W}) = 1, S = 0)$$

Propublica: *Separation (Equality of Odds)*

- ▶ Rather focuses on whether a person who does not re-offend is not wrongly predicted as being at high risk
- ▶ This score should be the same whatever the ethnicity

$$\mathbb{P}(f(\mathbf{W}) = 1 \mid Y = 0, S = 1) \approx \mathbb{P}(f(\mathbf{W}) = 1 \mid Y = 0, S = 0)$$

Impossibility result

Independence and separation

- ▶ Assume that f satisfies independence and sufficiency
- ▶ Then $Y \perp\!\!\!\perp S$

Independence and separation ($Y \in \{0, 1\}$)

- ▶ Assume that f satisfies independence and separation
- ▶ Then either $Y \perp\!\!\!\perp S$ or $f(\mathbf{W}) \perp\!\!\!\perp Y$ or both

Sufficiency and separation ($Y \in \{0, 1\}$)

- ▶ Assume that f satisfies sufficiency and separation
- ▶ Then either $Y \perp\!\!\!\perp S$ or $f(\mathbf{W}) \perp\!\!\!\perp Y$ or $\mathbb{P}(f(\mathbf{W}) = 1 | Y = 1) = 0$

🔗 In general only one of the constraints could be achieved

Fairness in classification: *Group fairness*

Setting

- ▶ Observations $\mathbf{W} = (\mathbf{X}, S)$ or $\mathbf{W} = \mathbf{X}$, and $Y \in \{0, 1\}$,
- ▶ Classifier $f \mapsto$ prediction $f(\mathbf{W})$

Definition of fairness

- ▶ **Demographic parity** (DP), for each $s \in \mathcal{S}$

$$\mathbb{P}(f(\mathbf{W}) = 1 | S = s) = \mathbb{P}(f(\mathbf{W}) = 1)$$

Example: Same loan approval rates across genders

- ▶ **Equalized odds** (EOd), for each $s \in \mathcal{S}$, and $y \in \{0, 1\}$

$$\mathbb{P}(f(\mathbf{W}) = y | S = s, Y = y) = \mathbb{P}(f(\mathbf{W}) = y | Y = y)$$

- ▶ **Equal opportunity** (EO), for each $s \in \mathcal{S}$

$$\mathbb{P}(f(\mathbf{W}) = 1 | S = s, Y = 1) = \mathbb{P}(f(\mathbf{W}) = 1 | Y = 1)$$

Example: Similar hiring rates for equally qualified candidates

Main approaches to enforce fairness

Pre-processing

- ▶ Find a feature representation $z \mapsto \phi(z)$
- ▶ $\phi(Z)$ independent on S

In-processing

- ▶ Based on the empirical risk minimization
- ▶ Given a set of predictor $\mathcal{F}$, solve

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{\mathcal{R}}(f) + \lambda \hat{C}(f),$$

with $\hat{\mathcal{R}}(f)$ empirical risk and $\hat{C}(f)$ empirical fairness constraints

Post-processing

- ▶ Given a pre-trained predictor f , not necessary fair
- ▶ Find a transformation $\hat{T}$ s.t. $\hat{T}(f)$ satisfies a desired fairness constraint

Takeaway message

Bias is everywhere!

- ▶ Historical bias in data because of social past behaviors
- ▶ Algorithms may amplify bias (acquisition/selection)
- ▶ Unethical conclusion
- ▶ All types of applications

Regulation at EU level

- ▶ Fast development on IA in high-risk applications
- ▶ Decision-making systems need regulation

Fairness

- ▶ Importance of understanding where is bias
- ▶ Several fairness paradigm
 - ▶ Individual fairness
 - ▶ Group fairness
- ▶ Several fairness definition
 - ▶ Demographic Parity
 - ▶ Equalized Odds / Equality of Opportunity
 - ▶ Sufficiency

Notion of fairness: *individual fairness*

“*treat like cases as like*” ($\leq$ Aristotle)

“*Ensure that similar individuals are treated similarly*”

C. Dwork et al., *Fairness through awareness*, (2012)

Example. Consider binary classification problem, where observations are of the form $(\mathbf{x}, y) \in \mathcal{X} \times \{0, 1\}$. Individual fairness *always* considers randomized predictions $f : \mathcal{X} \rightarrow \Delta(\{0, 1\})$

1. **Similarity of predictions:** $D : \Delta(\{0, 1\}) \times \Delta(\{0, 1\}) \rightarrow \mathbb{R}_+$

2. **Similarity of individuals:** $d : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_+$

Definition. A prediction $f : \mathcal{X} \rightarrow \Delta(\{0, 1\})$ is called *perfectly* (D, d) -individually fair if $\forall \mathbf{x}_1, \mathbf{x}_2$

$$D(f(\mathbf{x}_1), f(\mathbf{x}_2)) \leq d(\mathbf{x}_1, \mathbf{x}_2) .$$

Definition. A prediction $f : \mathcal{X} \rightarrow \Delta(\{0, 1\})$ is called *approximately* (D, d, α, γ) -individually fair if

$$\mathbf{P}_{(\mathbf{X}_1, \mathbf{X}_2)} (D(f(\mathbf{X}_1), f(\mathbf{X}_2)) > d(\mathbf{X}_1, \mathbf{X}_2) + \gamma) \leq \alpha ,$$

where $\mathbf{X}_1, \mathbf{X}_2$ are independent copies of $\mathbf{X}$.